25900525 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Value Addition Course (VAC) - MG3VACMBG200 - MICROBIAL PRODUCTS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minitus **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **2** marks

1. Mention how plant-based milks benefits people with lactose intolerance ?. [K] / [CO3]
2. Name each of a natural and artificial sweetner? [K] / [CO1]
3. What are skin probiotics? [K] / [CO2]
4. Name any two common sources of prebiotics. [K] / [CO2]
5. Define microbial nutrition. [K] / [CO1]
6. Define hemicellulose. [K] / [CO1]
7. What makes plant based eggs suitable for vegan? [K] / [CO3]
8. What is skinceuticals? [K] / [CO2]
9. Name two antimicrobial compounds obtained from fermentation. [U] / [CO1]
10. Which microorganism converts ethanol to acetic acid in vinegar. [K] / [CO1]
11. Write one health benefit of drinking kefir. [K] / [CO1]

12. Summarize the key features that define cosmeceuticals. [U] / [CO2]

[2x10 = 20]

Part B

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

13. Write the application of skinceuticals in any two skin conditions with example. [K] / [CO2]

14. Compare mango wine with grape wine. [U] / [CO1]

15. Explain the most effective ways to educate the public about the benefits and safety of microbial foods? [U] / [CO3]

16. Explain the role of probiotics in improving skin health and preventing skin disorders. [U] / [CO2]

17. Explain the economic & production challenges in acceptance of microbial foods. [U] / [CO3]

18. Describe any two value added Fermented Food Products. [K] / [CO1]

[5x4 = 20]

Part C

Long Essay Questions

Answer any **1** questions

Each question carries **10** marks

19. Define nutraceuticals. Describe their different categories with examples. [K] / [CO2]

20. Describe the role of friendly bacteria in the preparation and health benefits of kanji. [K] / [CO1]

[10x1 = 10]